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SETI

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Time Limit: 1 s Memory Limit: 64 MB
Submission: 13 AC: 2 Score: 99.87

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Input

On the first line of the input there is a single positive integer N , telling the number of test cases to follow. Each case consists of one line containing the value of p to use during the transcription of the string, followed by the actual string to be transcribed. The only allowed characters in the string are the lower case letters 'a'..'z' and '*' (asterisk). No string will be longer than 70 characters.

Output

For each transcribed string, output a line with the corresponding list of integers, separated by space, with each integer given in the order of ascending values of i .

Samples

input

[Copy](#)

```
3
31 aaa
37 abc
29 hello*earth
```

output

[Copy](#)

```
1 0 0
0 1 0
8 13 9 13 4 27 18 10 12 24 15
```

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Server Time: 2025-8-26 15:00:42